

Unsafe Meta-Strength

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In this paper, I show that by using a theorem grading of the HOOO EP axioms divided into 4 levels, the semantics of exponential propositions as function pointers using the Curry-Howard correspondence, starts to crack in level 3 and collapses in level 4. The cracks start to form in level 3 when there is a logic equal or stronger to Existential Logic, it makes negative propositions the same as paradoxical propositions. The collapse in level 4 is due to this level of meta-strength implying that Constructive Logic becomes an Existential Logic, hence fulfilling the conditions for the cracks to form on its own. The philosophical implications of this discovery are severe, since some statements about the properties of function pointers that seem self-evident, are in fact unsafe. I also apply these philosophical implications to discuss omniscience and quantum superpositions.

One of the most important proofs in mathematics and computer science all of time, was Gödel's work on the incompleteness theorems, that showed under a notion of provability given by Provability Logic, that theories containing arithmetic of natural numbers with addition and multiplication are either incomplete or inconsistent. Mathematicians have since explored the direction of the possibility mathematics being incomplete. Vladimir Voedvodsky, a Field medalist, expressed the concern that maybe mathematics is inconsistent. As a Path Semanticist, I hold the position that mathematics is both inconsistent and incomplete, but that we can design functional theories that are subsets of mathematics without these undesirable properties. However, I now also start to think that there is something basically wrong with logic in relation to the idea that it is possible to know everything, the property of omni-science, which is a popular ideal in mathematics.

Just think a little about this thought experiment: If I know A as a fact, then I might shape my beliefs around the ontology of A in the examples I find around me in reality. Now, if I learn B as a fact, then some of my beliefs around the ontology of A might change, but still, these beliefs are expected to have some well-grounded semantics such that I can explain why reality appeared to me in a such way that, when I first shaped my beliefs of A, I was mistaken. However, what if A and B both look reasonable in isolation, but when put together, then they make seemingly non-sensical predictions?

This such situation means, from a philosophical perspective, that it is impossible to know everything. We call the property of knowing everything for "omniscience". Up until Gödel, most mathematicians believed that it was possible to be omniscient, in principle. After Gödel published his work on the incompleteness theorems, mathematicians started suspecting that, at some point of increasing state of knowledge, things start to break down and collapse. With other words, the A and B that I used in the thought experiment, look reasonable in isolation, but do not go together.

While omniscience as a whole might be impossible, there is a difference in ways omniscience fails to be achieved. If things spiral out of control, such that there is an infinity of infinities, that not even any infinitely powerful mind in principle can know everything, then this mode of failure still feels a bit comfortable, in the sense that one can always think of a more powerful mind that knows more than any powerful mind one has imagined so far. Just like there is always a larger number, there is always a more powerful mind. However, to state that there is just one tangible powerful mind more powerful than any other mind, is absurd, because once you define how this powerful mind works in detail, you can always show that there is another mind that knows more. We call this property "incompleteness". Things spiral out of control, such that knowing everything becomes impossible. Most mathematicians like this property, because it simultaneously debunks some superstitious beliefs in the super-natural, while at the same time making more and more knowledge possible. So, while omniscience as an ideal is no longer worshiped, there is a preference in how it fails to be true.

The alternative to incompleteness is inconsistency. Inconsistency in a strong sense means that a theory can prove `false`. When this happens, it is possible to make any statement and claim it is true, no matter how ridiculous it sounds. Absurdness is a technical property in the language of logic, where any sentence you can construct that makes a proposition, is possible to construct. To guard ourselves from inconsistency, logicians design languages of logic such that absurdness is impossible or unsafe. When it is impossible, there is no situation where the user of that language can come into danger. When it is unsafe, the user gets a warning sign by the tool that they are about to do something risky. Inconsistency is not always a complete catastrophe for the language designer, but it is worth exploring these frontiers to better learn how to design proper tools.

Some people are born into this world with a naive outlook and self-confidence, believing that they can ignore all warning signs they see, which might benefit them in the short run, but they often end up dead or critically harmed. These kind of people do not understand why somebody else are warning them, because they often see these warnings as attempt to fool or brainwash others. For example, they see a sign “WARNING! ALLIGATORS” by a pond as somebody trying to keep other people out of the pond, because they want the whole pond for themselves. Thus, warning signs as part of a tool can produce a strange reaction in some people, because they have an immature mind.

In some cases, some people want inconsistency on purpose, because they feel that the absurdness they get as property of their thinking, allows them to do whatever they want to do. They see inconsistency as an open door, instead of something limiting. These people do not care about knowledge as such, but about goals they want to achieve. A tool that has a flaw in it, can sometimes be exploited to do something fun to somebody. For example, a computer with a flaw in its security system, might be hacked and controlled remotely without the owner knowing it. Whether the motivation is greed or negative social control, inconsistency can be used to achieve something.

Where incompleteness has a kind of idealistic desirable property for some people, that take a constructive and creative approach to knowledge, it can be difficult in design to achieve this sort of failure mode to be omniscient. Most of the time, when an error happens, it leads to inconsistency or to a weak inconsistency. A weak inconsistency is when the theory is no longer able to distinguish between two inputs that are supposed to be separated. There are many ways that weak inconsistencies can occur in complex systems, so preventing them by design is a tough challenge.

Again, because there are some people who ignore all warning signs and some people who seek out inconsistency on purpose, the person who's task is to achieve incompleteness and not inconsistency by design, is kind like Atlas, the person that carries the entire world on his back. If that one person fails at his job, then the whole world might literally collapse and burn, because there are so many idiotic humans that either destroys the world around them because they are not careful, or because they believe that can get something out of it, e.g. by forcing their deities to recreate the world. A lie that some people tell themselves and others, because they can not say they just want to destroy.

The thing is that some people just want to destroy. That is the truth. And even if these people rationalize their actions, whatever excuse they use will not matter, because if you just treat them as people that want to destroy, then the consequences would be the same. Here, the notion of “want” is not necessarily the same as an inner motivation, but a descriptive statement of their behavior.

Hence, in order to save the world and its people from destroying themselves, there is only system design that is powerful enough to stop it and this system design is only possible using mathematics and in turn, the mathematics shows that for any non-trivial system, it is impossible to achieve omniscience, but the best one can do is either incompleteness or inconsistency. Here, incompleteness is the most stable property of the two. However, if inconsistency is completely taken away, then it is kind of like taking people's toys: They get angry, whether it is justified or not.

On this intellectual journey toward designing systems for a world where people have a lot of freedom, but not so much freedom that they destroy themselves and the world, there are thousands upon thousands of buried people, that once tried to take the lives of the intellectuals, but failed.

This is the lesson of Atlas: He might look tired where he is standing, with the world on his shoulders, but you should probably not try to wrestle him in a death match.

Intellectually, there are giants and there are wasps. The wasps have no idea of what the giants are thinking. You might have noticed that a wasp does not behave in a way that shows it sees you as a human being. The wasp does not have the intelligence to make this distinction. To the wasp, a human being is part of a larger set of objects with similar attributes, most of them harmless. This is why it is so close that it is unaware that its life is in danger nearby human beings. A wasp might sting in defense, but to a human being this is often not fatal. The wasp on the other hand, might easily die in a such fight. You are a human being, but compared to Atlas, you are just a wasp.

The myth of Atlas is not told because the moral of the story is that you ought to not worry about the world because it is too heavy to lift, so better shrug it off. The moral is that if you attack somebody who is intellectually superior to you and threaten their lives, then you will soon find that you are not really destined to rule the world and the universe has not chosen to have that destiny. Some people see Atlas as somebody who carries the world because he wants it all for himself. Some people see Atlas as somebody it would be fun to provoke, because they want to watch how the world gets destroyed when Atlas loses control over it. Neither of them know what Atlas is thinking about.

The strength of Atlas is not muscles. To a scientist, the strength of muscles is relative to their function. There might be many other designs that are better for some tasks. The strength of Atlas is intellect. And when you do not know what Atlas is thinking about, you do not have anything to bring to the table. That is what it takes. Do you have something real to bring with you? Evidence?

Many people think about evidence as something abstract, but evidence is simply what makes something real. You do not eat food in an abstract way do you? The whole point with food is that is real. This idea is what evidence means. So, if you want to know what Atlas is thinking about, then you need to join the conversation by the table he sits. In order to do that, you need to bring something with you, evidence, or you will not get a seat by the table. These are the rules of table.

What would be interesting for Atlas to know, so that you could get a seat by the table he sits?

One such thing that Atlas might like to know, is how to construct a very simple system that has some weak inconsistency in it, that you can turn on and off using a knob. It should be a system that tells something about the nature of logic and by extension, mathematical knowledge. Of course, a system can not be designed using a random idea you came up with, because this makes it less likely to be relevant for mathematics in general. The weak inconsistency in it needs to appear from assumptions that look convincingly fundamental, solid and safe on the first impression.

To Atlas, the value of the system and the knowledge it could bring, is not in a childish sense that a system is either correct or wrong, with nothing in between. Details are important. A such system is not something to be thrown out in the garbage can because it has an error in it. On the contrary, it is the very nature of this error that makes it worth bringing to the table. It might surprise you what people bring, who get a seat by the table, and all the people who do not get a seat that brought something that looked promising, or something they valued because most other people around them do. A lot of ideas are trash in the disguise. You should not bring trash to the table where Atlas sits. If you do not have something to bring, then it is a good idea to keep a safe distance, but also sometimes, people think they have something, when it is trash, and laugh of those who get a seat.

Again, when a human being understands that a wasp is making a mistake, it does not bother them. Why would it? Human beings understand that human beings are human beings and wasps are wasps, which you can easily tell from their size and how they behave. However, to a wasp, it might recognize another wasp, but it is intellectually impossible for it to comprehend a human being.

Atlas is not an actual human being. What the people who use Atlas as an example try to say, is that there is something bigger than any individual human being. We use mathematics to design systems with properties that are intentional. These systems help us to keep people and the world safe. In relation between human beings and the complexity of the mathematics needed, our minds are closer to a wasp in that comparison. Maybe a very smart wasp can understand what “skin” means, but it would still be very far from comprehending what a human being is in its totality. Likewise, we rely on mathematical knowledge in ways that is impossible for our minds to comprehend the whole, but we use tools to do the job we need to do. Together, as a collective unit of intellectuals, we might understand the whole, but each individual might not have that ability. When we reason about the whole, we try to figure out how system design and knowledge go together.

This means, if it is possible to construct a system where there is a such weak inconsistency that can be turned on and off, this might help us understand more, collectively, about how system design and knowledge go together in the real world. Now you are starting to think like Atlas.

Logicians use Classical Logic a lot. Classical Logic is a language where propositions are either true or false. This does not mean that you can prove the proposition is true or you can prove the proposition is false. What you can prove, is that the proposition is either true or false.

This property might seem counter-intuitive: If something is either true or false, then it feels like it ought to be a way to prove it is true, or a way to prove it is false. Because, we are saying that it is either true or false and that is precisely what we mean by saying it. This holds when there is only one possible world. In reality, when we say something is real or unreal, there is usually evidence that shows it is real, or evidence that shows it is unreal. So, this intuition that there is a way to prove it is true, or a way to prove it is false, makes completely sense.

However, a language does not need to model just one possible world. In Logic, when we say that something is either true or false, it might be true in one possible world and false in another possible world. In every possible world, the thing is either true or false. This is how Classical Logic works. If we reduce Classical Logic to just one possible world, then there is a way to prove it is true, or a way to prove it is false. That is how evidence works in reality, because since reality is just reality, there is no other possible reality except in our imagination or in a different timeline which is not observable from our perspective as observers, we can use evidence as proof of something is real.

The problem is, when we allow multiple possible worlds, evidence is no longer proof. Evidence in this sense narrows the set of possible worlds you are in, but you have no proof what is real, except what the evidence says in isolation, plus the logical implications that follow. It is like, the world around you is shrouded into clouds and fog. You can see some part of the world, but you can not see it all at once. So, the possibilities that are beyond your observable horizon, makes thinking about multiple possible worlds meaningful. This is why Atlas got his head in the clouds: He does not know what is real and what is unreal, but he knows only some of it and he is thinking about it. But you, as a tiny human being, got no idea what Atlas is thinking about. You might have some clues, but your ability to grasp the bigger picture is non-existent. That is why you need to think carefully about what it means to have one's head in the clouds and what multiple possible worlds mean.

In Classical Logic, a proposition is either true or false in every possible world. There are no constraints on the number of possible worlds, so we do not know how many of them: One or many.

When we prove that something is true for all possible worlds, it follows that it must be true in any single possible world. In general, we think about Classical Logic as having many possible worlds, because it is more practical and useful. It is easier to go from many possible worlds to one, than the other way. Something that is true for all possible worlds is called a “tautology” and something that is false for all possible worlds is called a “paradox”. Now, if you have zero possible worlds, then tautologies and paradoxes collapse into the same idea. Because, something can be true in all possible worlds, but also false in all possible worlds, even it is true or false in all possible worlds, in the special case when there are zero possible worlds. With other words, you can not always tell a tautology from a paradox, but you might think you can, because you are only used to think about a set of possible worlds that is one or many. When you try to think about zero possible worlds, your experience and intuition fails you. You might end up somewhere, lost, not knowing where you are. Unlike Atlas. There is a set of possible worlds such that Atlas does not need to carry the world.

When Atlas does not carry the world, you have no idea where you are, because as a human being, you are depending on living on a planet. Atlas on the other hand, does not need to live on a planet and could have done something else with his time. However, if Atlas does not carry the world, then you are no longer alive and therefore you can not know how it is to think like Atlas. Although Atlas carries the world right now, Atlas knows what it is like to not carry it, but to you, that idea is just unimaginable. From your perspective, Atlas carries the world in every possible world. From Atlas’ perspective, he can distinguish between two sets of possible worlds, one set where he carries it and one set where he does not. To human beings, this is a profound mystery, similar to how a wasp is incapable of thinking about what it means to be a human being.

When we have a set of possible worlds, we can think about an empty subset with zero possible worlds, but we have no language to talk about how it is like to exist within that empty subset. When Atlas no longer carries the world, this is an empty subset to us, but not to Atlas. We can represent the state of Atlas not carrying the world using the empty subset, but there is no way for us to think about nuances within that state. It is an absurd world, from our perspective, where tautologies are no different from paradoxes. To talk meaningfully about some world, we need to differ tautologies from paradoxes, as a minimum starting point. This is the movement from zero possible worlds to at least one possible world. However, in just one possible world, there is no way to distinguish a tautology from a true proposition, or a paradox from a false proposition. This is the movement of one possible world to many.

Tautologies and paradoxes are even more important than propositions like `a` or `!a` in general, because they give meaning to something that can be known, as long there is at least one possible world. When we say `a`, we do not know how big the set of possible worlds is that one can say “`a` is true”. Likewise, when we say `!a`, we do not know how big the set of possible worlds is that one can say “`a` is false”. When we say `true`, this is a tautology. By saying it, we do not mean that something is true. We just say, `true`, as in general, a statement of truth. The only way to think of it as something is true, is when `true` is true. It turns out that in any possible world, you can say “`true` is true”. However, if we say that, then it sounds confusing, so it is better to just say `true`.

When we say `true`, we do not explain precisely what we mean. In some sense, the meaning behind the statement is at the same time trivial and clearly defined, but also entirely non-trivial and ambiguous. The reason for this ambiguity is because there can be multiple logical languages, where `true` takes on different meaning, depending on which language you are currently using. The same thing holds for `false`, which is a paradox. When we say “`false` is false” this holds in every possible world, so the statement is true, but the statement `false` in isolation can not be true. Therefore, when we make a judgement of a statement, that something is true or false, this is not the same thing as stating the statement. The statements `true` and `false` are tautology and paradox respectively, but not because of something within logic. Tarski calls this “truth correspondence”.

Tarski's idea of truth correspondence is that when we say a sentence `T`, we mean to make the judgement “`T` is true”. For example, the sentence “snow is white”, is true, because snow is white. However, when we say `false`, which is a paradox, we are not meaning “`false` is true”, because, it is self-evident that `false` can not be true. What Tarski actually talks about, is that statements refer to judgements in a particular context, so that we can talk about `true` and `false` inside and outside that context. This is how we know what `false` means, because we reason about it outside the context where it is stated as something that is true. Otherwise, we would not know that saying `false` inside this context is absurd.

What truth correspondence implies, is that when we learn the meaning of expressions, we are comparing contexts up against each other. For example, when we say “it is sunny”, we are comparing the context of that statement up against situations when we say “it is raining”. This is how we form meaning of words. Without comparison of contexts, distinctions between the meanings of words would become impossible. Thus, there has to be some distinction being made for a word to have meaning, but usually we do not know how this distinction was learned. We just learn words from our parents without knowing where the language of our parents came from. To Atlas, who has been around for as long the world has existed, he would laugh of the idea that you learn words without knowing where the words come from. Silly, tiny, human being.

This means, to impress Atlas, it is not sufficient to just present words that you have learned somewhere. For example, the word “God” impresses many human beings, but not Atlas. Because, you do not know where the word “God” comes from. You do not know what kind of meanings it had or what distinctions people made when they used this word originally. This is a typical trash idea in disguise: There is an error there, but the error is not even particularly intellectual interesting.

You might start to feel the intellectual difference between Atlas' mind and your own mind now. Perhaps you thought that just because something impressed you greatly, that it would surely impress Atlas? But no. What can you bring to the table, when it is so hard to come up with ideas, that is even worth thinking about, for such powerful minds like Atlas?

One such idea is how to express tautologies and paradoxes in Logic. Tautologies and paradoxes are constructs which theory belongs to meta-logic. Most mathematicians in their daily work use Sequent Calculus to do meta-logic where some language is modeled as expressions in Sequent Calculus, because, they gave up on the idea of uniting the object-language and meta-language, after thousands of years of thinking about it. Philosophers have tried since the Classical Period in Ancient Greece, to find a solution to this problem. In recent years it was discovered that it is possible to express tautologies and paradoxes in Logic using a bridge to computer programs. This bridge is the Curry-Howard correspondence. A tautology `a` is a function pointer of the type `true → a`. It is not a lambda `true => a`, because a lambda can capture variables from the environment, something a function pointer can not. When we say `true => a`, we do not know what set of statement we need to prove `a`. However, when we have a function pointer `true → a`, we know that `a` must be provable from `true` and nothing else. Of course, you can use axioms, which can be used everywhere. A paradox `a` is a function pointer of the type `a → false`. Therefore, meta-logic would be a set of axioms describing the properties of function pointers: The HOOO EP axioms.

HOOO EP stands for Higher Order Operator Overloading Exponential Propositions. The name comes from an idea of using higher order operator overloading of lambda expressions as a starting point. It turns out that this starting point is too strong, but after 5 months of reducing them, we got:

```
pow_lift : a^b -> (a^b)^c
tauto_hooo_imply : (a => b)^c -> (a^c => b^c)^true
tauto_hooo_or : (a | b)^c -> (a^c | b^c)^true
```

This means, the problem of uniting object- and meta-logic was finally solved. Meta-theorem proving of Constructive Logic was finalized. The work is done. Time to go home? Not yet.

The problem is Classical Logic. While you can go from Constructive Logic to Classical Logic by adding the axiom of excluded middle, which says that propositions are either true or false, the semantics of possible worlds breaks down and collapses when you use all 3 HOOO EP axioms. The semantics we get in the other end, is one where tautologies are the same as propositions being true and paradoxes are the same as propositions being false. This is a single possible world. Like reality.

Now, as I said earlier, it is good that in reality, we can know that evidence is proof of something being real, because reality is just reality and can not be many possible realities. However, to logicians, this is not good design, because they want to reason about many possible worlds as it is more practical and useful. The same goes for Atlas that got his head in the clouds.

First, let me explain the rationale behind the 3 axioms of HOOO EP, so you can see they are true:

`pow_lift : a^b -> (a^b)^c`

A type of a function pointer can be written two ways, `i → o` or `o^i`. So, `pow_lift` has the type to the right of `:`. This type is a function pointer, but the input and output are also function pointers. When we take in a function pointer of type `a^b` or `b → a`, we can produce a function pointer of type `(a^b)^c` or `c → (b → a)`. It means, a function is something you declare somewhere in your program and when you refer to it, you use a function pointer. Since you can access the function everywhere, it follows that if it has type `a → b`, it should be possible to declare some function type that takes `c` and returns `a → b`, since we can use the function pointer to `a → b` inside it.

The `pow_lift` axiom does not say what functions you have in your program. It says that, given some set of functions you have declared, you can construct a larger set of functions that use the functions you have already declared. It is like, the `pow_lift` axiom reasons as if the programmer writes every possible function when restricting himself to a starting code base. Every program he can construct, he creates, but not programs that only become possible by introducing some new axiom. In one sense the programmer is limited that humans would not like, but in another sense unlimited in sheer capacity. Just like Atlas is limited in the way that he carries the world by being in that role, but also unlimited in his capacity to do that task. Atlas might have his head in the clouds, but he has a very, very powerful mind to make up for this deficiency.

`tauto_hooo_imply : (a => b)^c -> (a^c => b^c)^true`

The next axiom, `tauto_hooo_imply`, says that if a function takes `c` and returns a lambda `a => b`, it is possible to create a function that takes `true` and gives us a lambda with input `c → a` and output `c → b`. The idea is, if we got `a` in the context `c`, we can produce `b` in the context `c`, because we already got `c` and hence can get `a => b`, but we need a way to get `a` also, the piece that is missing to get `b`. That missing piece is `c → a`. So, this axiom tells us it is possible to compose new functions where additional pieces of knowledge can produce even more knowledge.

The two axioms `pow_lift` and `tauto_hooo_imply` are so powerful together, that they can be used to prove most interesting theorems in meta-logic. Such theorems hold in both Constructive and Classical Logic, for one possible world and for many possible worlds. One might think that with two such powerful axioms, there would be no need for one more. However, since the construct is function pointers in programming, there is one more axiom needed to describe how they behave. It is the last axiom, `tauto_hooo_or`, that will cause all the deep philosophical implications, worries and concerns later on. It can not just be left out, because it would be intellectually dishonest.

Programming is naturally a constructive process. The logic that we use for computer programs, is Constructive Logic. When we use the Curry-Howard correspondence to do logic theorem proving by constructing computer programs, we are kind of hacking the language used for computer programs to achieve it. Therefore, as we add the 3rd axiom, ``tauto_hooo_or``, we are not betraying Logic, but staying faithful to Programming. Logic here is like a lover, but Programming is like the marriage partner. As in marriage, if the partner knows enough, then the possible-worlds semantics can collapse from many to one: The marriage. Programming gets first-priority.

$$\text{tauto_hooo_or} : (a \mid b)^c \rightarrow (a^c \mid b^c)^{\text{true}}$$

The axiom ``tauto_hooo_or`` says that when we got a function pointer ``c → a | b``, the type ``a | b`` is a sum-type. Meaning, it combines two types ``a, b`` and gives you two variants, ``left`` and ``right``, where ``left : a → a | b`` and ``right : b → a | b``. Naturally, when there are just these two ways to construct a type ``a | b`` by using ``left`` and ``right``, there must exist a way to get ``c → a`` and ``c → b``. It is physically impossible to get that type out in any other way.

There is no deep philosophical problem directly behind this axiom. Except, that there is.

If I have a function that takes ``a | b``, then I got an element of the combined sum-type of ``a, b``, but I do not know whether that element was constructed using ``left`` or ``right``. Which means, if I have a function ``f : a | b → a``, I can always use ``x == f(left(x))`` to convert back and forth. If I have a function ``g : a | b → b``, I can always use ``y == g(right(y))`` to convert back and forth. Since ``x : a`` or ``y : b`` must exist somewhere, there is always some way to construct at least one of the types ``a, b``. However, I do not know this from the argument ``a | b`` alone. What ``tauto_hooo_or`` does is to give us a way to say that these two possibilities are the only ones.

In Logic, this does not hold up, because if even if ``a | b`` is true, this does not mean that there is a way to prove ``a`` or a way to prove ``b``. It would only be true if there was only one possible world.

This is where Programming and Logic disagree. In Programming, there is just one possible world. The type checker is not checking the source of many possible computer programs. It just checks one concrete source for a computer program. In Logic, we want there to be many possible worlds, because this is more practical and useful. It is kind of like when people argue over whether a marriage ought to be open or closed. Logic wants the marriage to be open, so love is not constrained to a single marriage, while Programming wants the marriage to be closed, limiting love to marriage.

In Path Semantics, there is no final “answer” that we can give people. There are just logical consequences from the choices of design we make. In the same way there are consequences either way, whether people choose an open or closed marriage. However, if you have a very strong opinion about marriage and try to bring it to the table where Atlas sits, he will not be impressed.

You can try to reason your way out of this problem, but it is doomed to fail from the start. In one sense, it is very practical and useful to think about many possible worlds, but when it comes down to it, all that you can ever experience is just one possible world: Reality.

For example, you might think about some possible world where you can wrestle with Atlas in a death match and win, but this possible world is probably not the one you live in. Perhaps you get in a few punches, but it might not be sure that Atlas even notices, before he accidentally crushes you. There are usually two reasons why people end up in this situation: When they ignore all warning signs and take huge risks, or they see inconsistency as an opportunity to do what they want. They might want to live, but they also want to fight Atlas. Their problem is that when they see Atlas carrying the whole world, they see a vulnerable, tired, giant, not an inhumanly strong wrestler.

You should have taken the hint. A giant that carries the entire world, is not somebody you should mess with, even the giant looks tired to you, because you have no idea what it is thinking. Perhaps the giant looks tired, because of something it was thinking about at the moment when you saw it. So, it could be that it does not look tired because of the weight of the entire world. The myth of Atlas is designed in a such way to bait those people who want to destroy, into revealing their true selves. It is part of one of the many systems that keep people and the world safe.

Do you see how Atlas makes the myth of Atlas? Atlas is a metaphor for the collective intellectual capacity that people have, to design a world that enable and empower people's freedom, but at the same time prevent people who want to destroy from doing what they want. The myth creates another myth, of its own interpretation, using the first myth as tool that reveal who you truly are by the way you react to it psychologically. This is the level of difference in intellect, between a wasp and human being.

It is this myth of Atlas that I use to explain to you what unsafe levels of meta-strengths are. So, I am using a self-referential myth to explain another concept that might be interesting to talk about with Atlas, by the table where he sits. Which is yet another myth. This is an example of meta-reasoning.

Now, it is not just a challenge to understand the basic idea of unsafe levels of meta-strength, but also, we need to dive into the details. We need to find the knob where we can turn on and off the weak inconsistency.

As a starting point, one can use Classical Logic, which has an axiom of excluded middle:

$$\text{excm} : \text{true} \rightarrow a \mid !a$$

There is a weaker logic than Classical Logic, called "Existential Logic", where there is an axiom:

$$\text{excm_ex} : \text{true} \rightarrow !a \mid !!a$$

All Classical Logics are Existential Logics. All Existential Logics are Constructive Logics.

The reason we use Existential Logic here, is because this is our knob. When it is on, we can see at what level of meta-strength is needed to enter the unsafe territory, where Programming starts to differ from Logic. To measure meta-strength, we use a theorem grading:

$$\text{grade meta_strength} [\text{pow_lift}, \text{tauto_hooo_imply}, \text{tauto_hooo_tauto_or}, \text{tauto_hooo_or}];$$

When any theorem uses `pow\_lift`, it gets the grade 1 or higher in meta-strength. When any theorem uses `tauto\_hooo\_or`, it has grade 4, which is the highest in meta-strength. When none of these axioms are used, they get the grade 0 and are not listed. The axiom `tauto\_hooo\_imply` has grade 2. We introduce a new axiom, `tauto\_hooo\_tauto\_or`, which is a stronger version of `tauto\_hooo\_or` where `c = true`, but otherwise the same.

It turns out that when there is an Existential Logic and one uses grade 3 in meta-strength, it is possible to prove that `!a`, a negative proposition, can be turned into `false^a`, a paradox. The same is not possible for a tautology, but one can turn `a`, a positive proposition, into `(!a)^true`, a tautology of the double-negated proposition.

The form `(!a)^true` known to prove the theorem `a` in Classical Logic from within Constructive Logic. Including excluded middle, so `(!!(a | !a))^true` is provable. This means, any proposition `a` must have a way to be constructed to be real, in a classical sense, without making any assumptions.

From a programming perspective, this makes sense. If we are limited to constructs of functions, then there is no way to run the program except by using `true` as input, because there is no input or output in this language. Because we are running the program using `true` as the only input, any constructed state must necessarily follow from `true`.

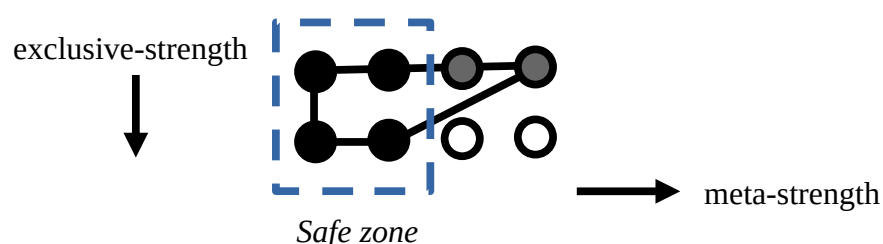
Similarly, in a brute force theorem proving model of classical logic, we give a proof a list of parameters, but there are no assumptions built in, except the ones that the user adds as part of the theorem they want to prove. Therefore, the evaluation process of the proof also follows from making no assumptions, with can be thought of as starting with `true` and nothing else.

The reason $\neg a$ can be turned into false^a , is because false^a is the same as $(\neg\neg a)^{\text{true}}$. This means, the collapse of paradoxes and negative propositions is for the same reason that any evaluated proof must follow from `true` in a classical sense, meaning that the states of the program are determined.

Now, from a logical perspective, since Existential Logic causes problems at meta-strength 3 for many possible-worlds semantics, this is the lowest unsafe level in general. If the user does not use Existential Logic, then what happens at meta-strength 4? What are the conditions needed to cause unsafe collapse there?

It turns out, that at meta-strength 4, Constructive Logic becomes an Existential Logic. So, there are no further conditions needed to trigger the unsafe collapse. Like before, the collapse is not complete, since only negative propositions can be turned into paradoxes. One needs a final step, to add the axiom of the excluded middle, to turn positive propositions into tautologies.

The philosophical implications of this discovery are profound. It means, in the absence of meta-logic, we are fine whether we use Constructive or Classical Logic. However, when adding meta-logic, we need to be careful about how close we are to Classical Logic. This means, meta-logic as a whole is naturally a constructive process. In Classical Logic, the full strength of meta-logic collapses, so that the meta-language and object-language become one of the same thing. One can also think about this as the axiom of excluded middle adding a 5th level of meta-strength. In practice though, it is beneficial to separate the two, because when we work with Existential Logic, we might want to use 0 meta-strength. We divide the two up into two gradings, one for meta-strength with 4 levels and one for exclusive-strength with 2 levels (Existential and Classical). This way, we use two gradings which can be thought of as a boat with a safe zone and an unsafe zone:



The closer you are to the tip of the boat, the more careful you should be. Here, a black node means that it is safe. A gray node means there is a partial collapse. A white node means full collapse.

Since what we know about computer programs eliminates many possible-worlds semantics, it is becomes impossible to know everything in the sense that you can know everything about Programming and everything about Logic at the same time. You might find a way to organize your everyday knowledge into two overlapping domains, but in principle, when we talk philosophically about the property omniscience, it is not possible make it work. There is a tradeoff between different kinds of knowledge and they can not be known at the same time in the same sense.

Now, there is this idea that in quantum physics, a physical system can be in superposition of multiple states at the same time. If we can prove using the double-slit experiment, that an electron goes through one slit or the other, but not both, then the interference pattern vanishes. However, if we can not prove which of the two slits the electron goes through, we see an interference pattern. This is kind of like using the axiom of excluded middle. We do not choose which slit the electron goes through, but we force it to be one or the other. However, in superposition, the electron might be thought of as going through both slits at once, because it can be in more than one state. What is interesting about this experiment is that the more we know about the location of the electron, the less we know about the many possible-worlds semantics that can infer from the interference pattern when repeating the experiment many times. So, just like increasing meta-strength collapses many possible-worlds semantics, we observe a similar phenomena in the double-slit experiment.

Perhaps this is what happens in the Everett interpretation of quantum mechanics? Quantum mechanics is a language we use, that might be thought of as a high abstraction. When theoretical physicists are reasoning about quantum gravity, they try to figure out what the lower abstractions of this language might be. One class of theories is that particles are made up of tiny strings. However, at the smallest scales that are measurable in physics, at Planck scales, some people believe that reality can be described using hypergraph rewriting systems, that correspond to Constructive Logic. With other words, Constructive Logic could be the language our reality is written in. As information gets entangled with the environment over time, observers measure determined states in their experiments. We know the system can be in superposition, but all measurements we make are determined values. The behavior of an electron can be changed, depending on what kind of information we measure about it. When we see this measurement in a bigger picture that includes the observer, there is information that both determines how the measurement is made and what is measured. This information causes the “collapse” of the superposition into determined outcomes. The system moves from a many possible-worlds semantics, to a single possible world.

In the Everett interpretation, as many possible-worlds semantics move into single possible-world semantics, this is because an observer can no longer interact with some of the previous possible worlds. The observer splits into different Everett branches, each making different measurements. Like when we use an empty set of possible worlds to represent when Atlas is no longer carrying the world, we can not reason about the other previous possible worlds we no longer can interact with, but this does not mean that there are none observers who can. Atlas can reason just fine about not carrying the world, but you can not, because you would not be alive to learn what might happen. This is the basic idea, that many possible-worlds semantics can have dynamics over time and evolve into states where observers can use evidence as proof of something being real or unreal.

However, be aware: This high level of meta-strength is unsafe. Be cautious.